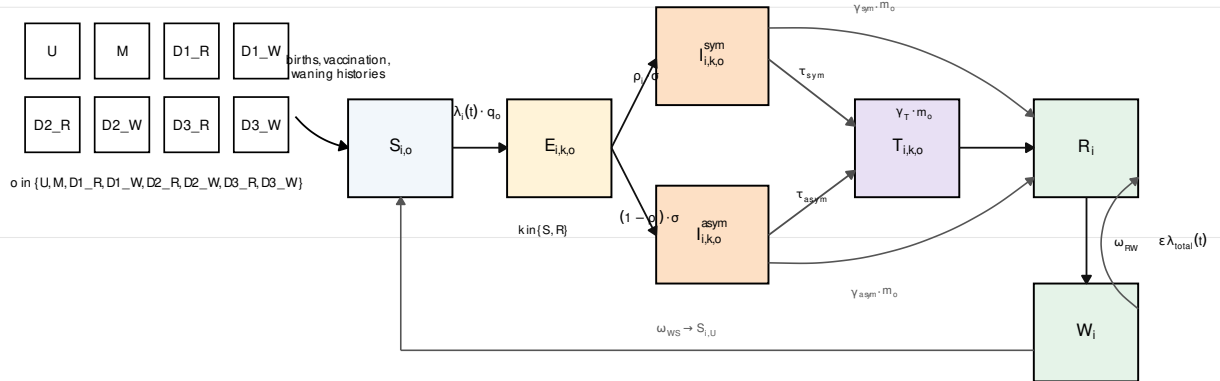


A

Pertussis SIRWS transmission schematic

One age group shown; the ODE repeats this 74-compartment template across 8 age groups coupled by C_{ij} .

Susceptible histories



Infectious pressure: $\eta_{a,o} (I^{sym} + r_A I^{asym} + \zeta_{a,kT})$; PEP modifies $\lambda_{ik}(t)$.

Demographic turnover, importation, seasonal NPI forcing, vaccination maintenance and resistance anchoring enter as modifiers or external flows.

B